

 Marwadi University	Marwadi University Faculty of Technology Department of Information and Communication Technology	
Subject: Artificial Intelligence (01CT0616)	Aim: To obtain the classification of various classes using k-Nearest Neighbor approach	
Experiment No: 03	Date:	Enrollment No:92410133034

Aim: To obtain the classification of various classes using k-Nearest Neighbor approach

IDE: Google Colab

Theory:

This algorithm is used to solve the classification model problems. K-nearest neighbor or K-NN algorithm basically creates an imaginary boundary to classify the data. When new data points come in, the algorithm will try to predict that to the nearest of the boundary line. Therefore, larger k value means smother curves of separation resulting in less complex models. Whereas, smaller k value tends to over fit the data and resulting in complex models. It's very important to have the right k-value when analyzing the dataset to avoid over fitting and under fitting of the dataset.

The model representation for KNN is the entire training dataset. It is as simple as that. KNN has no model other than storing the entire dataset, so there is no learning required. Efficient implementations can store the data using complex data structures like k-d trees to make look-up and matching of new patterns during prediction efficient. Because the entire training dataset is stored, you may want to think carefully about the consistency of your training data. It might be a good idea to curate it, update it often as new data becomes available and remove erroneous and outlier data.

K-NN algorithm assumes the similarity between the new case/data and available cases and put the new case into the category that is most similar to the available categories. K-NN algorithm stores all the available data and classifies a new data point based on the similarity. This means when new data appears then it can be easily classified into a well suite category by using K- NN algorithm. K-NN algorithm can be used for Regression as well as for Classification but mostly it is used for the Classification problems.

K-NN is a non-parametric algorithm, which means it does not make any assumption on underlying data. It is also called a lazy learner algorithm because it does not learn from the training set immediately instead it stores the dataset and at the time of classification, it performs an action on the dataset. KNN algorithm at the training phase just stores the dataset and when it gets new data, then it classifies that data into a category that is much similar to the new data.

Example: Suppose, we have an image of a creature that looks similar to cat and dog, but we want to know either it is a cat or dog. So for this identification, we can use the KNN algorithm, as it works on a similarity measure. Our KNN model will find the similar features of the new data set to the cats and dogs images and based on the most similar features it will put it in either cat or dog category.

Pre Lab Exercise:

- What is K in KNN-approach?

K represents the **number of nearest neighbors** used to classify a new data point.

The algorithm checks the K closest data points and assigns the class based on majority voting.

A small K can lead to overfitting, while a large K can smooth the decision boundary.

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b. Which are the types of distance metrics?

Common distance measures used in KNN are:

- **Euclidean Distance** (most common)
- **Manhattan Distance**
- **Minkowski Distance**
- **Hamming Distance** (for categorical data)

These help in measuring similarity between data points.

c. What can be the criteria of selection of the value of K?

- K should not be too small (avoid overfitting)
- K should not be too large (avoid underfitting)
- Usually choose **odd value of K** (to avoid ties)
- Best K is selected using **trial and error or cross-validation**

d. What are the advantages of KNN algorithm?

- Simple and easy to understand
- No training phase (lazy learner)
- Works for both classification and regression
- Good for small datasets

Methodology:

1. Load the basic libraries and packages

```
[20]
✓ 0s  import numpy as np
      import pandas as pd
      import matplotlib.pyplot as plt|
```

2. Load the dataset

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```
[21]
✓ 0s # Create simple dataset for classification

data = pd.DataFrame({
    'X1': [1, 2, 3, 6, 7, 8],
    'X2': [2, 3, 1, 7, 8, 6],
    'Label': [0, 0, 0, 1, 1, 1]
})

data
```

	X1	X2	Label
0	1	2	0
1	2	3	0
2	3	1	0
3	6	7	1
4	7	8	1
5	8	6	1

Next steps: [Generate code with data](#) [New interactive sheet](#)

3. Analyse the dataset

```
[24]
✓ 0s # Analyse dataset
data.describe()
```

	X1	X2	Label
count	6.000000	6.000000	6.000000
mean	4.500000	4.500000	0.500000
std	2.880972	2.880972	0.547723
min	1.000000	1.000000	0.000000
25%	2.250000	2.250000	0.000000
50%	4.500000	4.500000	0.500000
75%	6.750000	6.750000	1.000000
max	8.000000	8.000000	1.000000

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4. Pre-process the data

```
[25]
✓ 0s
# Check missing values
data.isnull().sum()
```

...

0

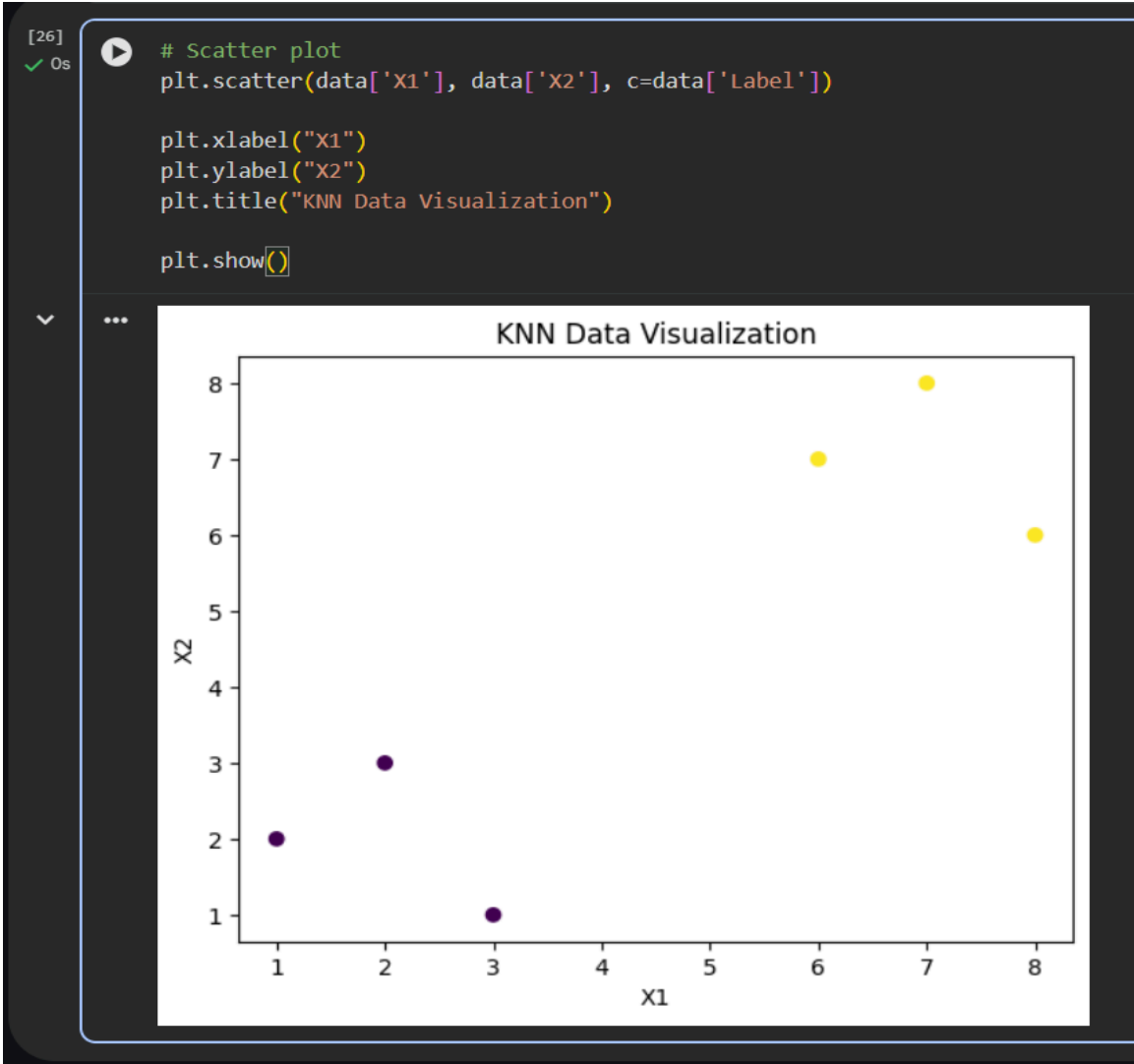
X1 0

X2 0

Label 0

dtype: int64

5. Visualize the Data



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6. Separate the feature and prediction value columns

```
[27]
✓ 0s # Separate features and label

X = data[['X1', 'X2']].values
y = data['Label'].values

X, y

... (array([[1, 2],
            [2, 3],
            [3, 1],
            [6, 7],
            [7, 8],
            [8, 6]]),
      array([0, 0, 0, 1, 1, 1]))
```

7. Select the number K of the neighbors

```
[28]
✓ 0s # Select number of neighbors

K = 3

K

... 3
```

8. Calculate the Euclidean distance of K number of neighbors

```
[29]
✓ 0s # Euclidean distance function

def euclidean_distance(p1, p2):
    return np.sqrt(np.sum((p1 - p2)**2))
```

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9. Take the K nearest neighbors as per the calculated Euclidean distance.

```
[30]
✓ Os
# New data point to classify
new_point = np.array([5, 5])

# Calculate distances
distances = []

for i in range(len(X)):
    dist = euclidean_distance(new_point, X[i])
    distances.append((dist, y[i]))

# Sort distances
distances = sorted(distances)

# Get K nearest neighbors
neighbors = distances[:K]

neighbors

... [(np.float64(2.23606797749979), np.int64(1)),
      (np.float64(3.1622776601683795), np.int64(1)),
      (np.float64(3.605551275463989), np.int64(0))]
```

10. Among these k neighbors, count the number of the data points in each category.

```
[31]
✓ Os
# Count labels of neighbors
labels = [label for _, label in neighbors]

# Final prediction
prediction = max(set(labels), key=labels.count)

prediction

... np.int64(1)
```

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11. Assign the new data points to that category for which the number of the neighbor is maximum.
the new data point is assigned to the class which has the maximum number of nearest neighbors (majority voting).
12. Our model is ready.
After classification, the KNN model is ready to predict new unseen data.

Results:

To be attached with

- a. Confusion matrix
Confusion matrix is used to evaluate the classification performance by comparing actual and predicted values.
- b. Accuracy Score
Accuracy score shows the percentage of correct predictions made by the KNN model.

Observation and Result Analysis:

- a. Nature of the dataset
Dataset contains two classes with clear separation based on feature values.
- b. During Training Process
KNN does not train a model; it stores the dataset and calculates distances during prediction.
- c. After the training Process
Model correctly classifies new data based on nearest neighbors.
- d. Observation over the Learning Curve
Model accuracy depends on the value of K; proper K gives better results.

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Post Lab Exercise:

- a. Is KNN useful in regression-based problems?
Yes, KNN can be used for regression by taking the average of the values of K nearest neighbors.
- b. Why KNN is a non-parametric algorithm?
KNN is non-parametric because it does not assume any fixed form or distribution of the data.
- c. What are the assumptions of KNN approach?
KNN assumes that similar data points are located close to each other in feature space.
- d. How can the KNN approach make the prediction of the unseen dataset?
KNN predicts new data by finding K nearest neighbors and assigning the class based on majority voting.
- e. Why is KNN called a Lazy Learner?
KNN is called a lazy learner because it does not train a model and directly uses stored data during prediction.
- f. Why should we not use KNN for large dataset?
KNN is slow for large datasets because it calculates distance from all data points, increasing computation time.